

Algebraic Topology

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1 Boundary Operator

Let X be a space and $Y \subset X$ a subset. The set $\partial_X Y$ is defined as

$$\overline{Y} - Y^\circ$$

where the closure and interior are taken with respect to space X , and this is known as the boundary of Y . I may write ∂Y if the ambient space was explicitly stated or where it could cause no confusion.

For spaces X, Y , $A \subset X$, $B \subset Y$, we have the following identity:

$$\partial(A \times B) = A \times \partial_Y B \cup \partial_X A \times B$$

Proof: TODO. Of particular interest is that this is of the same form as the product rule in calculus.

2 Discs and Spheres

Let D^n be the equivalence class (when not stated explicitly, assume the equivalence relation to be homeomorphicity) of a closed ball in $\mathbb{R}^n$. Then $S^{n-1} = \partial D^n$.

3 Cell Complexes

Let the set X^0 be known as the 0-skeleton and be a set of points. For $n > 0$,

$$X^n = (\{D_i^n\}, \phi)$$

where $\{D_i^n\}$ is some set of n -discs and $\phi : S^{n-1} \rightarrow X^{n-1}$ maps the boundary of then n -skeleton to the $n - 1$ -skeleton.

4 Euler Characteristic

The Euler characteristic χX of the space X given by the cell complex $(X^0, X^1, \dots, X^n)$ is defined to be

$$\sum_{i=0}^n (-1)^i |X^i|$$

where $|X^i|$ is the number of i -discs in the i -skeleton. If $X \cong Y$, then $\chi X = \chi Y$,
proof: TODO.